

Coding Audit Summary (Easy Read)

Date: 2026-02-18

Project: NecessityEntrepreneurs

Source of truth: .lyx paper

What this is:

A plain-language summary of coding issues that could affect results.

Top 3 risks:

- 1) Entrepreneur output is zeroed in simulation in two places.
- 2) Education index is used before assignment in key loops.
- 3) Matching object mismatch in vacancy/search cost (p vs q).

Why these matter

Risk 1 (output zero): can bias GDP/output moments and any ratio using output.

Risk 2 (education index): can feed wrong policy function by education, which can distort entrepreneurship, unemployment, and welfare statistics.

Risk 3 (p vs q): changes effective hiring/search costs and firm size choices.

Bottom line:

If risks 1 and 2 were active in runs, headline quantitative results may move.

Recommended immediate actions

- 1) Patch output formula in both simulation blocks.
- 2) Patch education index assignment order + CEV indexing.
- 3) Fix GE convergence criterion for theta.
- 4) Decide and document $p(\theta)$ vs $q(\theta)$ in code and paper.
- 5) Re-run baseline and compare core tables/figures.

Prepared for coauthor handoff:

- CODE_REFEREE_REPORT.md
- PAPER_CODE_CROSSWALK.csv
- FIX_CHECKLIST.md